

1. Administrative & Study Identification

Full Study Title: Observational Study of the Effectiveness and Safety of the MEK 1/2 Inhibitor (MEKi) Koselugo (Selumetinib) in Pediatric Patients with Symptomatic, Inoperable, Neurofibromatosis Type 1 (NF1)-Associated Plexiform Neurofibromas (PN)
Short Title/Acronym: PEDIA Study
Protocol Number: PEDIA **NCT Number:** NCT06175637
Sponsor Name: AstraZeneca
Investigational Product: Selumetinib (Koselugo)
Phase of Development: N/A (Observational Study)
Medical Monitor: AstraZeneca Clinical Medical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the real-world effectiveness and safety of selumetinib for Chinese pediatric patients with NF1-associated Plexiform Neurofibromas (PN).
Secondary Objectives: To expand the understanding of disease characteristics and treatment patterns of NF1-PN in a real-world setting, and to evaluate patient-reported outcomes including pain evolution.

2.2 Background & Rationale To address the data gap regarding the effectiveness and safety of selumetinib for Chinese pediatric NF1-PN patients. Neurofibromatosis type 1 (NF1) is a rare genetic disorder, and Plexiform Neurofibromas (PN) affect 30%-50% of these patients, causing pain, functional impairment, and disfigurement. Following the recent approval of selumetinib by the China National Medical Products Administration (NMPA) for pediatric patients ≥ 3 years with symptomatic, inoperable PN, this study aims to generate real-world evidence in the Chinese population to guide clinical practice.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 80 to 100 eligible pediatric patients **Number of Sites:** Multicenter (China) **Estimated Study Start:** 20-Dec-2023 **Estimated Primary Completion:** 30-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Pediatric patients aged 3 ≤ age ≤ 16 years old at the time of study enrollment. 2. Diagnosed with NF1 as per the revised 2021 Guidelines. 3. Have symptomatic, inoperable Plexiform Neurofibromas (PN). 4. Intend to use selumetinib after enrollment. 5. Patient or patient's parent/legal guardian must be willing and able to give written informed consent.

4.2 Exclusion Criteria 1. Evidence of malignant peripheral nerve sheath tumor (MPNST), prior malignancy, or other cancer requiring treatment with chemotherapy or radiation therapy. 2. A life-threatening illness, medical condition, or organ system dysfunction. 3. Have had prior treatment with a MEK inhibitor, Ras, or Raf inhibitor. 4. Patients currently participating in any clinical trials at the time of enrollment or initiation of selumetinib.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral administration of selumetinib based on body surface area (BSA), conforming to the approved local labeling and clinical practice standards. Route of Administration: Oral (Capsules) Duration of Treatment: Chronic continuous daily dosing until disease progression, unacceptable toxicity, or treatment discontinuation.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures for pain and NF1-related morbidities. Prohibited: Concurrent use of other MEK, Ras, or Raf inhibitors, or participation in interventional clinical trials.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:--
:--	:--	Screening	Day -28 to 0 Informed Consent/ Assent, Medical History, Baseline MRI Baseline Day 0 Selumetinib Initiation, PROs scales assessment, Pain (NRS-11) Interim Visits Per Routine Care Safety monitoring, Dose modification tracking, Tumor volume assessment, Pain tracking End of Study Nov 2027 Final data collection, Exit Interview, Survival/progression status

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Real-world effectiveness characterized by the presence, number, and size (volume reduction) of PN, as well as overall safety and treatment

modification rates. **Measurement Method:** Clinical assessment, routine MRI evaluation, and patient medication diaries.

7.2 Secondary & Exploratory Endpoints 1. Time to tumor regrowth after treatment discontinuation. 2. The evolution of pain measured with the NRS-11 scale compared to baseline. 3. Treatment adherence and persistence (time from first dose to discontinuation date).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Method of collection includes patient diaries and routine healthcare provider (HCP) interviews during scheduled follow-up visits. **Serious Adverse Events (SAEs):** Reported continuously per standard real-world pharmacovigilance protocols (typically within 24 hours of awareness). **Data Monitoring Committee (DMC):** AstraZeneca internal safety monitoring team.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients who meet eligibility criteria and receive at least one dose of selumetinib in the real-world setting. **Sample Size Justification:** A target of 80-100 patients is deemed sufficient to provide robust descriptive statistics and fulfill data gaps regarding safety and effectiveness in the Chinese demographic. **Handling of Missing Data:** Persistence of treatment will be analyzed via survival analysis methods; course of disease post-discontinuation will be evaluated descriptively.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), local regulatory requirements (NMPA), and the Declaration of Helsinki. **Monitoring Plan:** Data collected prospectively through electronic Case Report Forms (eCRFs) reflecting routine clinical visits. **Audit Trail:** Robust eCRF locking mechanisms; secure data-sharing and patient privacy protection measures established by the sponsor.

11. Study Identifiers & Governance

Primary ID: PEDIA **Registry Number:** NCT06175637 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PEDIA **Official Regulatory Title:** Observational Study to Evaluate the Effect and Safety of Selumetinib in Pediatric Patients With NF1-PNs

Primary Condition: Neurofibromatosis Type 1 (NF1) **Diagnosis Threshold:** Clinically confirmed NF1 with symptomatic, inoperable Plexiform Neurofibromas **Medical Methodology:** MEK 1/2 Inhibitor Targeted Therapy **Optimization Target:** PN volume stabilization/reduction and morbidity (pain) relief

Management Pathway: Standard real-world clinical pathway for pediatric NF1-PN **Education Protocol (HCP):** Local investigator training on Real-World Evidence (RWE) collection and AE reporting **Education Protocol (Patient):** Guidance on accurate completion of medication diaries and NRS-11 PRO scales **Quality Audit Mechanism:** Medication diary compliance tracking and routine monitoring visits

Total Duration: Approximately 4 years (Dec 2023 to Nov 2027)
Onsite Visit Cadence: Aligned with standard local clinical practice for NF1-PN monitoring
Remote Monitoring Frequency: Supplemental via medication diary reviews
Checklist Review Interval: Periodically during standard of care check-ups

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				